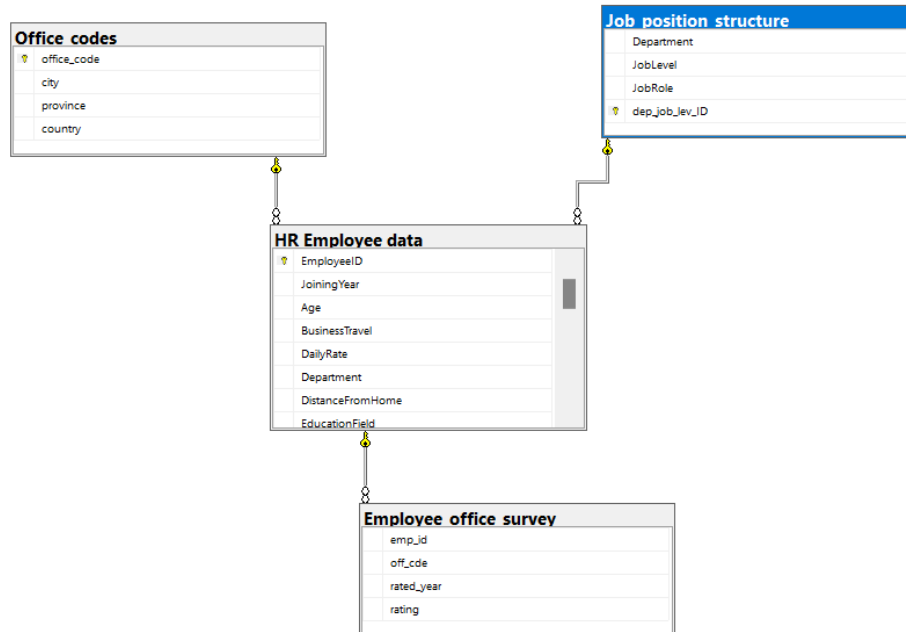


MODEL



-- 1. Find employees who have more than 5 years of experience and work overtime

SELECT

EmployeeID,

dep_job_lev_ID

FROM

[HR Employee data]

WHERE

TotalWorkingYears > 5 AND OverTime = 'Yes'

Results Messages

	EmployeeID	dep_job_lev_ID
1	100000	Corporate Functions L7
2	100001	Corporate Functions L6
3	100003	Corporate Functions L7
4	100004	Marketing L5
5	100005	Corporate Functions L7
6	100011	Corporate Functions L7
7	100013	Corporate Functions L6
8	100015	Corporate Functions L6
9	100017	Corporate Functions L6
10	100025	Corporate Functions L6
11	100028	Corporate Functions L6

-- 2. Show average rating per office for the last 3 years

WITH MaxYear AS (

SELECT MAX(rated_year) AS max_year

FROM Employee_office_survey

)

SELECT

oc.office_code,

AVG(eos.rating) AS avg_rating

FROM

Office_codes oc

LEFT JOIN [HR Employee data] hr

```

ON oc.office_code = hr.office_code
LEFT JOIN Employee_office_survey eos
ON hr.EmployeeID = eos.emp_id
JOIN MaxYear my
ON eos.rated_year BETWEEN my.max_year - 2 AND my.max_year
GROUP BY oc.office_code
ORDER BY AVG(eos.rating) DESC

```

	office_code	avg_rating
1	PHL	8.879769
2	OTT	8.874010
3	TOR	8.854716
4	MKM	8.851497
5	VAN	8.839592
6	CAL	8.805555
7	BOS	7.012902
8	NYC	7.005840

-- 3. Find employees with the highest JobSatisfaction and the lowest MonthlyIncome (lowest income up to + 100 dollars)

```

WITH Max_job_satisf AS (
    SELECT MAX(JobSatisfaction) AS MaxSatis
    FROM [HR Employee data]
),
MinIncomeValue AS (
    SELECT MIN(MonthlyIncome) AS MinInc

```

```
FROM [HR Employee data]
)
```

```
SELECT
```

```
    EmployeeID,
    JobSatisfaction,
    MonthlyIncome
```

```
FROM [HR Employee data], Max_job_satisf, MinIncomeValue
```

```
WHERE
```

```
    JobSatisfaction = MaxSatis
```

```
    AND MonthlyIncome BETWEEN MinInc AND (MinInc + 100)
```

Results Messages			
	EmployeeID	JobSatisfaction	MonthlyIncome
1	103946	4	1107.00
2	107768	4	1104.00
3	108581	4	1081.00
4	111244	4	1102.00
5	111712	4	1051.00
6	111743	4	1105.00

-- 4. Show the number of employees who left the company by reason for leaving

```
SELECT
```

```
    Reason,
```

```
    COUNT(EmployeeID) AS Number_of_Employees
```

```
FROM [HR Employee data]
```

```
WHERE Attrition = 'Yes'
```

```
GROUP BY Reason
```

```
ORDER BY Number_of_Employees DESC;
```

Results		Messages
	Reason	Number_of_Employees
1	retiring	312
2	medical issues	311
3	offered more money	300
4	return to school	297
5	long hours	296
6	reallocation out of the area	281
7	unhappy	281
8	career change	266
9	poor work performance	228
10	cost cutting	185
11	Layoffs	184

-- 5. How many employees are there in each combination of department and job role?

SELECT

jps.Department,

jps.JobRole,

COUNT(hr.EmployeeID) AS Number_of_Employees

FROM

Job_position_structure jps

JOIN

[HR Employee data] hr

ON jps.dep_job_lev_ID = hr.dep_job_lev_ID

WHERE

hr.Attrition = 'No'

GROUP BY

jps.Department,

jps.JobRole

ORDER BY

Number_of_Employees DESC;

Results Messages			
	Department	JobRole	Number_of_Employees
1	HR	HR Coordinator	681
2	Product	Product Coordinator	670
3	Delivery	Service Delivery Analyst	651
4	Sales	Account Executive	644
5	Marketing	Coordinator	640
6	Marketing	Specialist	636
7	Delivery	Senior Service Delivery Analyst	626
8	Corporate Functions	Administrative Coordinator	620
9	Corporate Functions	Administrative Assistant	618
10	HR	HR Generalist	614
11	Sales	Sales Development Representative	614

-- 6. Show the total number of employees by category of number of companies they worked for (categories to be created in the query)

SELECT

MAX([HR Employee data].NumCompaniesWorked) AS Max_num_companies_worked,

AVG([HR Employee data].NumCompaniesWorked) AS Avg_num_companies_worked,

MIN([HR Employee data].NumCompaniesWorked) AS Min_num_companies_worked

FROM

[HR Employee data]

Results Messages			
	Max_num_companies_worked	Avg_num_companies_worked	Min_num_companies_worked
1	9	2	0

SELECT

CASE

WHEN NumCompaniesWorked <= 2 THEN '0-2 companies before'

```

        WHEN NumCompaniesWorked BETWEEN 3 AND 5 THEN '3-5 companies before'

        WHEN NumCompaniesWorked > 5 THEN 'a lot of companies before'

    END AS CompaniesWorkedCategory,

    COUNT(EmployeeID) AS Number_of_employees

FROM

    [HR Employee data]

GROUP BY

    CASE

        WHEN NumCompaniesWorked <= 2 THEN '0-2 companies before'

        WHEN NumCompaniesWorked BETWEEN 3 AND 5 THEN '3-5 companies before'

        WHEN NumCompaniesWorked > 5 THEN 'a lot of companies before'

    END

ORDER BY Number_of_employees DESC;

```

Results Messages		
	CompaniesWorkedCategory	Number_of_employees
1	0-2 companies before	8324
2	3-5 companies before	3733
3	a lot of companies before	1366

-- 7. Show the number of employees by education level who received their last promotion within 5 years or less, and who have been with the company for less than 9 years

```

SELECT

    EducationField,

    COUNT(EmployeeID) AS Number_of_employees

FROM

    [HR Employee data]

WHERE

    YearsSinceLastPromotion <= 5 AND

```

YearsAtCompany < 9

GROUP BY

EducationField

ORDER BY

COUNT(EmployeeID) DESC

Results		Messages
	EducationField	Number_of_employees
1	Doctorate	3367
2	Diploma	3339
3	Masters	3253
4	Bachelors	3215

-- 8. Show the number of employees by department and country

SELECT

country,

Department,

COUNT(EmployeeID) AS Number_of_employees

FROM

[HR Employee data]

JOIN Employee_office_survey ON [HR Employee data].office_code =
Employee_office_survey.off_cde

JOIN Office_codes ON Employee_office_survey.off_cde = Office_codes.office_code

GROUP BY

country, Department

ORDER BY

COUNT(EmployeeID) DESC

Results Messages			
	country	Department	Number_of_employees
1	Canada	Corporate Functions	7968284
2	Canada	Sales	7570614
3	Canada	Product	7479863
4	Canada	Delivery	7453375
5	Canada	Marketing	7439537
6	Canada	HR	7400873
7	US	Sales	5908365
8	US	Product	5785517
9	US	Corporate Functions	5725079
10	US	Delivery	5665347
11	US	HR	5529586

-- 9. Show all employees who had JobSatisfaction below 3 and PerformanceRating below 3

SELECT

EmployeeID

FROM

[HR Employee data]

WHERE

JobSatisfaction < 3 AND

PerformanceRating < 3

Results Messages	
	EmployeeID
1	100060
2	100081
3	100094
4	100127
5	100199
6	100211
7	100221
8	100259
9	100295
10	100405
11	100463
12	100488

-- 10. Show all offices that have at least one employee with a rating of 5

SELECT

 DISTINCT HighPerformers.office_code AS office_codes

FROM

 (SELECT

 *

 FROM

 [HR Employee data]

 WHERE

 PerformanceRating = 5) AS HighPerformers

JOIN Employee_office_survey ON HighPerformers.office_code =
Employee_office_survey.off_cde

JOIN Office_codes ON Employee_office_survey.off_cde = Office_codes.office_code

	office_codes	PerformanceRating
1	BOS	5
2	CAL	5
3	MKM	5
4	NYC	5
5	OTT	5
6	PHL	5
7	TOR	5
8	VAN	5

-- 11. Find employees who have OverTime = 'Yes' and WorkLifeBalance = 1

SELECT

EmployeeID,

OverTime,

WorkLifeBalance

FROM

[HR Employee data]

WHERE

OverTime = 'Yes' AND WorkLifeBalance = 1

	EmployeeID	OverTime	WorkLifeBalance
1	100110	Yes	1
2	100158	Yes	1
3	100249	Yes	1
4	100432	Yes	1
5	100513	Yes	1
6	100679	Yes	1
7	100739	Yes	1
8	100820	Yes	1
9	100831	Yes	1
10	100834	Yes	1
11	100964	Yes	1

-- 13. Show average age by department

```

SELECT
    jps.Department,
    AVG(hr.Age) as Average_Age
FROM
    Job_position_structure jps
JOIN
    [HR Employee data] hr
ON jps.dep_job_lev_ID = hr.dep_job_lev_ID
GROUP BY
    jps.Department
ORDER BY
    AVG(hr.Age) DESC

```

Results Messages		
	Department	Average_Age
1	Corporate Functions	37
2	Delivery	36
3	Sales	36
4	HR	35
5	Marketing	35
6	Product	35

-- 14. Show the top 5 highest paid employees per office

```

WITH CleanedData AS (
    SELECT DISTINCT
        eos.off_cde,
        hr.EmployeeID,
        hr.MonthlyIncome

```

```
FROM
    Employee_office_survey eos
JOIN
    [HR Employee data] hr
    ON eos.emp_id = hr.EmployeeID
),
```

```
Top5ByOffice AS (
    SELECT
        off_cde,
        EmployeeID,
        MonthlyIncome,
        ROW_NUMBER() OVER (
            PARTITION BY off_cde
            ORDER BY MonthlyIncome DESC
        ) AS RowNum
    FROM
        CleanedData
)
```

```
SELECT
    off_cde,
    EmployeeID,
    MonthlyIncome,
    RowNum
FROM
```

Top5ByOffice

WHERE

RowNum <= 5

ORDER BY

off_cde,

RowNum;

Results Messages				
	off_cde	EmployeeID	MonthlyIncome	RowNum
1	BOS	100059	19973.00	1
2	BOS	110307	19920.00	2
3	BOS	110912	19874.00	3
4	BOS	100065	19845.00	4
5	BOS	103327	19840.00	5
6	CAL	100375	19973.00	1
7	CAL	100364	19948.00	2
8	CAL	100740	19943.00	3
9	CAL	101415	19854.00	4
10	CAL	100198	19822.00	5
11	MKM	105095	19883.00	1

-- 15. Pivot the Employee_office_survey table so you can easily see the change of average rating per office year by year.

SELECT

off_cde,

AVG(CASE WHEN rated_year = 2017 THEN rating END) AS Year_2017_AvgRating,

AVG(CASE WHEN rated_year = 2018 THEN rating END) AS Year_2018_AvgRating,

AVG(CASE WHEN rated_year = 2019 THEN rating END) AS Year_2019_AvgRating,

AVG(CASE WHEN rated_year = 2020 THEN rating END) AS Year_2020_AvgRating,

AVG(CASE WHEN rated_year = 2021 THEN rating END) AS Year_2021_AvgRating,

AVG(CASE WHEN rated_year = 2022 THEN rating END) AS Year_2022_AvgRating,

AVG(rating) AS Average_rating_all_years

FROM

Employee_office_survey

GROUP BY

off_cde

Results		Messages						
	off_cde	Year_2017_AvgRating	Year_2018_AvgRating	Year_2019_AvgRating	Year_2020_AvgRating	Year_2021_AvgRating	Year_2022_AvgRating	Average_rating_all_years
1	NYC	7.026984	7.021582	6.967013	6.991004	7.013157	7.011169	7.002909
2	MKM	8.619631	8.586690	8.596604	8.627777	8.918489	8.995756	8.773846
3	OTT	8.625000	8.612139	8.700407	8.661378	8.942648	8.989294	8.811421
4	CAL	8.607336	8.668032	8.700110	8.565660	8.850132	9.000000	8.763525
5	TOR	8.565934	8.663106	8.644498	8.655679	8.913413	8.986285	8.791509
6	VAN	8.628813	8.750467	8.637724	8.628335	8.899258	8.971512	8.791522
7	PHL	8.608381	8.578671	8.639240	8.649886	8.968037	8.987238	8.803330
8	BOS	7.002976	6.964347	6.992485	7.018037	7.016726	7.004604	7.005336